MDM4U Final Exam

Name:

K/U	A	T	C	Total
24	20	22	22	88

K/U:

1. Evaluate [6 marks]
   a. 7!
   b. ${}_7P_1$
   c. .${}_7C_1$
   d. P(7, 7)
   e. $\binom{7}{2}$
   f. C(7, 2)

2. Use the binomial theorem to expand $(3x - 2y)^5$ [2 marks]

3. In how many ways can 12 different cars be parked in the front row of a used-car lot if the owner does not want the red one beside the orange one because the colours clash? [2 marks]

4. What is the probability that a random integer between 1 and 50, inclusive, is not a prime number? [2 marks]

5. If upper-case and lower-case letters are considered as different letters, how many six-letter computer passwords are possible [4 marks]
   a. With no repeated letters?
   b. With at least one capital letter?

6. How many divisors of 4725 are there? How many of these divisors are divisible by 5? [4 marks]

7. Eight friends, three of whom are left-handed, get together for a friendly game of volleyball. If they split into two teams randomly, what is the probability that one team is composed of
   a. All right-handed players?
   b. Two right-handed and two left-handed players? [4 marks]

Applications:

8. Adult salmon jumping over a series of logs as they swim upstream to spawn. The salmon have a 0.6 probability of a successful jump if they rest prior to the jump, but only a 0.3 probability immediately after jumping the previous log. If the fish are rested when they come to the first log, what is the probability that a salmon will clear
   a. Both of the first two logs on the first try without resting?
   b. All of the first four logs on the first try if it rests after the second jump? [4 marks]

9. The weather forecast calls for a 12% chance of rain tomorrow, but it is twice as likely that it will snow. What is the probability that it will neither rain nor snow tomorrow? [2 marks]

10. A lottery has sold 5000000 tickets at $1.00 each. The prizes are shown in the table. Determine the expected value per ticket. [2 marks]

prize	Number of prizes
$1000000	1
$50000	10
$500	100
$10	1000

11. Sasha and Pedro find that after winning a game, Sasha has a 65% probability of winning the next game. Similarly, Pedro has a 60% probability of winning after he has won a game. Pedro won the game last week. [6 marks]
    a. What are the probabilities of each player winning this week?
    b. What is the probability of Pedro winning the game two weeks later?
    c. If Pedro and Sasha play 100 games, how many games is each player likely to win?

12. The speed limit in a school zone is 40 km/h. a survey of cars passing the school shows that their speed are normally distributed with a mean of 38 km/h and a standard deviation of 6 km/h
    a. What percent of cars passing the school are speeding?
    b. If drivers receive speeding tickets for exceeding the posted speed limit by 10%, what is the probability that a driver passing the school will receive a ticket? [4 marks]

13. Determine the probability distribution for the number of heads that you could get if you flipped a coin seven times. [2 marks]

T/I:

14. Suppose that 82.5% of university students use a personal computer for their studies. If ten students are selected at random, what is
    a. The probability that exactly five use a personal computer?
    b. The probability that at least six use a personal computer?
    c. The expected value of students who use a personal computer? [6 marks]

15. The defective rate on the CD producing assembly line has gone up to 12%, and the department head wants to know the probability that a skid of 50 CD players will contain at least 4 defective units. Use the binomial distribution to answer this question, and uses a normal approximation. By what percent will binomial distribution answer exceed normal approximation? [4 marks]

16. A news paper poll indicated that 70% of Canadian adults were in favour of the antiterrorist legislation introduced in 2001. The poll is accurate within $\pm 4\%$, 19 times in 20. Estimate the number of people polled for this survey. [2 marks]

17. A box contains 15 red, 13 green, and 16 blue light bulbs. Bulbs are randomly selected from this box to replace all the bulbs in a string of 15 lights

    a. Estimate the expected number of each colour of light bulb is the string
    b. Calculate the theoretical probability of having exactly 5 red bulbs in the string
    c. What is the expected number of blue bulbs? [6 marks]

18. Aerial surveys of 50 randomly selected 100 km^2 sections of the park had a mean of 1.67 wolves and a standard deviation of 0.32 Determine a 95% confidence interval for the mean number of wolves per 100 km^2 in the park. [4 marks]

Communications:

19. a) What is a Bernoulli trial? (2marks)

b) What is the key difference between trials in a geometric probability distribution and those in a hypergeometric probability distribution? (8 marks)

20. Give an example of each of the following sampling techniques and list an advantage of using each type. (4 mark each total 12 marks)
    a) stratified sample
    b) simple random sample
    c) systematic sample